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Secure blockchain based intrusion detection for IoT networks

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Abstract

A blockchain-enabled Model integrates blockchain technology with Intrusion Detection Systems to enhance the security of Internet of Things (IoT) networks. It ensures data integrity, decentralization, and tamper-proof logging of intrusion detection. The approach improves trust, transparency, and real-time threat detection in distributed IoT environments. The existing blockchain-based IDS approaches, Blockchain Enabled (BCE-IoT), uniquely integrate blockchain consensus with federated-style local training, lightweight cryptography, and Shapley Additive Explanations (SHAP)-based explainability, ensuring both security and interpretability in IoT environments. The proposed work combines Blockchain technology with explainable artificial intelligence solutions to create a new cybersecurity Model that strengthens intrusion detection within IoT networks. The proposed model enhances transparency in tracking cyberattacks by combining blockchain security storage capabilities with SHAP, an explainable AI. This research utilises machine learning and artificial intelligence to detect threats in real-time, countering Distributed Denial of Service (DDoS), Denial of Service (DoS), scanning, Cross-Site Scripting (XSS), injection, password, and backdoor attacks. BCE-IoT delivers more precise security by combining blockchain's permanent data features and AI anomaly detectors, thereby reducing security alert mistakes. The performance effectiveness of Blockchain-Enabled IoT surpasses that of the Content Integrity Detection System. It combines Blockchain and Software-Defined Networking to enhance security in network environments, utilising blockchain-based mutual confirmation for software-defined networking to detect and block cyber threats. The evaluation establishes BCE-IoT as an effective IoT network security solution that delivers strong cybersecurity features, is adaptable to modern connected environments, and offers interpretable security solutions. The performance evaluations demonstrate that BCE-IoT provides a robust, flexible, and interpretable cybersecurity solution suitable for modern IoT environments.

Keywords Internet of things, Blockchain, Intrusion detection, Artificial intelligence, Software defined network, Distributed denial of service

1 Introduction

The interpretability solution provided by SHAP is well-suited for explaining machine learning models, especially in problems with multiple outputs, and requires an appropriate dataset. The SHAP theoretical approach describes the production of machine



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learning models. The essential component of NetFlow Telemetry and Operational Network IoT Version (NF-ToN-IoT-V2) determines how each feature contributes separately to detecting network traffic between malicious and benign categories, and security engagement with SHAP values to acquire contextual insights about their analysis. This is because the values reveal supporting interactions between features, which enhance model security and improve visibility. A breakdown of model classifications through the split in the class's technique enables the determination of SHAP values for individual features, including positive and negative SHAP values, as well as the magnitude for different attack types in NF-ToN-IoT-V2. Healthcare providers and cyber analysts can gain better control over intrusion detection systems, enhancing cybersecurity measures and improving the interpretability of threats. These abilities yield more trustworthy AI-based security solutions [1]. This study focuses on detecting multiple cyberattack scenarios present in the NF-ToN-IoT-V2 dataset, including Distributed Denial of Service (DDoS), Denial of Service (DoS), scanning, Cross-Site Scripting (XSS), injection, password attacks, backdoors, Man-in-the-Middle (MITM) attacks, and ransomware.

The organizations are now demanding more interpretive machine learning models for cybersecurity applications, which has led to the growing complexity of system threats. SHAP is an advanced interpretability technique that reveals to users the decision-making process of machine learning models. SHAP effects allow predictive outcome variables to become interpretable through SHAP analyses because the method applies game-theoretic concepts for distributing feature contributions towards predictions [2]. The multi-class split method enables SHAP to analyze multi-class classification data by providing separate SHAP value computations for every class, improving the interpretation of classification systems.

The construction of dependable interconnected systems depends on complete security measures with open visibility throughout the entire process in IoT operations. The combination of modern, explainable techniques, such as SHAP, with multi-class split boosts users' ability to understand their predictive models and the predictive knowledge they hold. NF-ToN-IoT-V2 data can be effectively utilised in an IoT network study; however, blockchain technology can ensure the authenticity of the data and its traceability. When these methods are used in combination with dual systems to enhance security credibility, security solutions of the distributed systems are better detected [3].

The NF-ToN-IoT-V2 data provides a detailed model to address contemporary network security needs related to intrusion detection issues. The current network traffic database comprises common attack and traffic patterns, which serve as useful metrics for assessing the success of the machine learning model. NF-ToN-IoT-V2 is analysed with the SHAP multi-class split method to identify attacks and normal network processes, and identify important features. In this way, security experts can develop more effective intrusion detection models, which can aid in selecting the most suitable detection features. The detection of security threats is based on four key features, including packet length, protocol, source and destination IP addresses, failed login attempts, and abnormal system calls. The models are more successful in threat detection because they analyse these key features [4]. SHAP multi-class split technique allows researchers to obtain true outcomes when analyzing the critical importance of global features by reducing errors in interpretation. The feature strength depends on the class separation of attacks adopted during the splitting. A specific feature of network flows was demonstrated

to identify Denial of Service (DOS) attacks using SHAP multi-class split methods, but it still struggled to detect data exfiltration [5]. A thorough study at the grade level is crucial for cybersecurity, as every type of attack has its own unique characteristics. Integrating NF-ToN-IoT-V2 enables the better interpretation of security solutions generated through AI systems, leading to improved system reliability. Security professionals achieve model authentication and accuracy optimization by rationally understanding their predictive logic [6]. Establishments implementing cybersecurity standards would benefit from explainable features because these capabilities enhance the quality of their decision-making and compliance with industry standards. The NF-ToN-IoT-V2 assessment with SHAP multi-class split is used to achieve a high level of interpretability, which improves detection performance and enables the documentation of machine learning-based intrusion detection systems. The analysis of per-attack SHAP value makes it possible to conduct research that helps to better understand decision patterns, which improves cybersecurity defences.

In this work, Blockchain is used not only as a tool of immutability but also to distribute consensus in combining local model updates, authenticating devices in the IoT, and recording intrusion incidents without tampering. Similarly, SHAP has been iteratively applied to multi-class intrusion detection, allowing for per-attack interpretability of feature contributions to DDoS, injection, and ransomware, and informing feature selection. These specific applications extend beyond ordinary blockchain and SHAP applications, directly enhancing the detection precision and reliability of BCE-IoT.

The effectiveness of the proposed BCE-IoT framework is evaluated using commonly used benchmarks, including accuracy, precision, recall, specificity, and F1-score, to provide a comprehensive assessment of the detection performance.

To address these limitations, BCE-IoT introduces three novel design aspects: (i) integration of blockchain consensus with federated-style local training for decentralized trust management, (ii) use of XXTEA is also known as Tiny Encryption Algorithm (TEA) TEA and Elliptic Curve Cryptography (ECC) to provide lightweight cryptographic security for IoT devices, and (iii) SHAP-based explainability for enhanced interpretability. These features collectively distinguish BCE-IoT from prior blockchain-enabled IDS frameworks.

The article's structure is organized as follows Sect. 2 presents the Literature Review, highlighting existing research related to intrusion detection and IoT security. Section 3 outlines the Existing Materials and Methods used in the study, including dataset selection and experimental setup. Section 4 introduces the Proposed Blockchain-Enabled Internet of Things Model, detailing its architecture and components. Section 5 presents the experimental results and provides an in-depth analysis of the model's performance. Section 6 concludes the article, outlining key findings and future research directions.

2 Literature review

Le et al. [7] developed an IDS that combined the integration of Decision Trees (DT) and a Random Forest (RF) classifier to increase its detection rate. The team utilised three IoT-based datasets, namely NF-BoT-IoT-v2, NF-ToN-IoT-v2, and IoTDS20, as research input and modelled transparency using SHAP methods. The accuracy of detection was significantly increased, as were precision, F1-scores, and recall values. The SHAP system gave users clear explanations that displayed both model decisions and the essential

characteristics that influenced the results. Through a combination of these techniques, scholars develop more effective IDS frameworks to address the security of the IoT and enhance their readability.

Chakir et al. [1] demonstrate that Machine Learning with Explainable Artificial Intelligence (XAI) enhances Web attack monitoring of Industrial Internet of Things (IIoT) security infrastructure, making it more transparent. The project team utilised SHAP to select features and gained a clear understanding of what their models do. Their strategy showed considerable changes in all web attack detection performance measures, as well as accuracy, precision, and recall performance measures. When explainable machine learning models are combined with SHAP technology, web attack detection systems in IIoT environments have superior performance results and interpretability capabilities. Muna et al. [8] assess the necessity of smart cities to possess attack detection systems capable of effectively protecting against attacks with explainable models; however, traditional deep learning approaches are not interpretable at all. The article utilises the IOTD20 data and XGBoost and XAI technologies to derive valid detection models and provide clear analysis. The study resulted in attack detection solutions that enhanced the massive IoT assault detection by increasing credible modelling with improved precision and recall values. Combining XGBoost with XAI presents a dependable and safe process for monitoring attacks within smart city IoT systems. Fraihat et al. [9] introduced a robust IDS development approach that addresses network traffic processing issues for secure IoT applications by utilising systems that manage large-scale network traffic. The system incorporates features from a modified version of the Arithmetic Optimization Algorithm (AOA) to perform feature selection, which enhances detection performance. Reducing NetFlow features through precise selection enhances the efficiency of intrusion detection algorithms that utilise Random Forest and Extra Trees, due to their improved operational capabilities. Applying the Arithmetic Optimization algorithm in large-scale IoT networks makes Machine learning algorithms more efficient and scalable. Chakraborty et al. [10] evaluate multi-objective, multi-label, and multi-class classification methods, and then apply them to healthcare, agriculture, entertainment, education, and finance to enhance decision-making capabilities. The authors emphasise the importance of establishing protective systems that balance technological development with personal privacy and data protection requirements. These classification methods revealed their ability to advance practical performance through ethical protocols. These modern technologies enable better decision-making and accountability in industries by combining ethical and secure operational practices.

Majhi et al. [11] enhanced Intrusion Detection Systems (IDS) through NF-UNSWNB15-v2 dataset analysis combined with FOX optimization, which employs V shaped transfer functions for obtaining features and Light-GBM classification. The Light-GBM model performs classification with improved detection of DoS, Exploits, and Fuzzers attacks and minimizes selected feature counts beyond previous standards. The reported results demonstrate enhanced accuracy, alongside precise and reliable detection, which proves the effectiveness of their technique. The combination of FOX optimization with Light-GBM creates an IDS that operates more effectively and precisely for network attack detection.

Amin et al. [12] showed how CZOLSTM represents an algorithm that unifies deep learning with swarm intelligence for handling Cyber Threat Detection problems. The

CSE-CIC-IDS2018 dataset enabled CZOLSTM to produce superior results in binary and multiclass classification compared to multiple advanced algorithms. The results demonstrate impressive improvements across accuracy, recall, F1 scores, and precision levels. Combining chaotic zebra optimization and LSTM enables cyber threat detection methods to achieve superior accuracy and operational efficiency in their system operations. Mutalib et al. [13] describe how traditional IDS encounter challenges with APT detection because they produce substantial incorrect alarms alongside substandard recognition accuracy. The authors developed APT detection models that perform better and provide higher interpretability through deep learning models enhanced by XAI methods, SHAP, and LIME. The study shows that XAI and improved interpretability capabilities improve APT detection performance in model development. To develop trustworthy cybersecurity solutions for APT detection, XAI must be a foundation. Medaram et al. [14] introduce malware detection with mitigation mechanisms that defend cloud-based environments from security problems resulting from adaptive shared cloud infrastructure. The research solution uses signature detection, ML anomaly detection systems, blockchain security features, and Behavioral analysis to optimize cloud security. The authors confirm through research findings that real-time threat monitoring, virtualization security, and encryption technology successfully fight adaptive malware threats. Integrating protected AI systems offers two key benefits: enhanced cloud resilience and solutions to internal threats and security issues associated with multi-cloud infrastructure.

Javed et al. [15] established a compact intrusion detection system that protects basic IoT equipment in smart home networks. The system uses machine learning algorithms to conduct real-time attack detection that requires minimal computations, thus fixing network issues and saving power. The study establishes the system as providing good IoT device security, great rates of detection, and low rates of erroneous alert generation. The IDS based on machine learning is an effective and valuable tool to protect IoT devices within smart homes. Hernandez-Jaimes et al. [16] introduce LSH with machine learning functionality that is used to detect infrequent patterns in IoT data to enable effective detection of anomalies in IoT networks. In order to maintain real-time functionality, the approach is based on resource-effective operations, which are feasible on limited IoT systems. The system recorded improved detection results when operating at high speeds, as well as features that enabled it to adjust to different conditions. The quality of detection performance, as well as performance improvement, makes the LSH-based model a viable IoT network security solution. As an example of IoT-based maritime system protection, Nkoro et al. [17] propose the Zero-Trust Architecture (ZTA), which is based on Explainable Artificial Intelligence (XAI) as a primary threat detection tool to enhance system protection while maintaining greater openness.

The model involves continuous monitoring and dynamic access control, and addresses maritime communication network vulnerability and unauthorised access in real-time. It is a stronger model utilized on systems and improves information exchanges and information accuracy in identifying. Zero Trust Architecture (ZTA) combined with Explainable Artificial Intelligence (XAI) represents a significant improvement in IoT marine cybersecurity, as it integrates explainable detection with preventive defensive operations. Yao et al. [18] proposed an information protection model for biometric privacy in UAV delivery systems, utilising edge computing and sophisticated encryption approaches. Its architecture enables the safe storage and processing of biometric data at

the edge without any latency, reducing cloud dependencies. Their design enhances user authentication and reduces threats of data leakage when conducting UAV operations. Yao et al. [19] proposed a privacy-sensitive framework for collecting location information in intelligent systems deployed in edge environments. The system can protect confidential geolocation data through safe aggregation and lightweight cryptography, and can be used to analyse the data. The solution enables a good balance to be struck between data utility and stringent privacy specifications. Dong et al. [20] presented a blockchain-aided Self-Sovereign Identity (SSI) framework to enhance security in UAV delivery systems with edge computing. The proposed architecture decentralizes identity management, eliminating reliance on third-party providers. Their study demonstrated improved authentication, traceability, and user privacy in UAV logistics. Yao et al. [21] developed a security framework for edge-enabled UAV delivery systems, addressing vulnerabilities in data transmission and access control. The model integrates encryption, trust management, and secure communication protocols to safeguard UAV operations. Experimental results confirm its efficiency in mitigating attacks while maintaining system scalability.

Ahanger et al. [22] discuss IoT-specialized intrusion detection systems, which address resource limitation weaknesses and architectural diversity. They compare AI-oriented solutions, including machine learning, deep learning, and federated learning, on Host-Based IDS (HIDS), Network-Based IDS (NIDS), and hybrid setups, and detect threats with high accuracy, using fewer resources and offering more flexibility. The paper also explores IoT-focused attacks, including Sybil, malicious node and DoS attacks, and it shows that AI-assisted IDS models can successfully address sophisticated intrusion trends in limited IoT settings.

Despite these advancements, current research on intrusion detection still has significant limitations. The majority of them are built on centralised architectures and hence they bring about single points of failure as well as scaling issues. Some of these models have high accuracy but cannot be interpreted to have in-still confidence in actual IoT security implementations. Furthermore, previous studies often utilise datasets with a limited number of attack examples, and recent threats, including ransomware and MITM, are not adequately represented. There are not many frameworks that combine blockchain to decentralize trust or use explainability to ensure transparency. The identification of these gaps underscores the relevance of the proposed BCE-IoT model, which combines blockchain consensus and lightweight cryptography with SHAP-based explainability to address such issues.

In summary, although previous IDS methods, including ensemble trees with SHAP, Fractional Order Exponential (FOX) LightGBM, and Chaotic Zebra Optimisation with Long Short-Term Memory (CZOLSTM) demonstrated high accuracy, they have several problems, including computational overhead, reliance on Centralized trust, and low interpretability. To fill these gaps, the proposed BCE-IoT will combine blockchain consensus, lightweight encryption, XXTEA, ECC, and SHAP-based explainability, thereby turning the solver into a more powerful and open IoT IDS.

3 Existing materials and methods

This section describes the SHAP multiclass split NF-ToN-IoT-V2 dataset as a training and testing resource. It also introduces blockchain and machine learning models that support the proposed BCE-IoT systems. In addition to standard evaluation metrics,

SHAP were used to interpret the predictions of the ML classifiers, providing feature-level importance for intrusion detection.

3.1 NF-ToN-IoT-V2 dataset

The NF-ToN-IoT-V2 database contains 43 attributes focusing on packet size elements that researchers use to analyze network traffic patterns. Among the metrics are Total Length of Forward and Backward Packets and Max, Min, Mean, and Standard Deviation measurements for both forward and backward packet lengths. A thorough analysis of network flow packet size distribution is derived from a set of features that include Max Packet Length, Min Packet Length, Packet Length Mean, Packet Length Standard Deviation, and Packet Length Variance. Network intrusion detection systems heavily rely on packet size features because they enable the system to identify regular traffic from malicious traffic patterns. The data completeness is evident in this dataset through the absence of missing values, while most features are distinct, except for one replicated description. Decentralized architecture ensures superior security and trust because it stays immune to tampering activities and fraud schemes. Blockchain technology supports cybersecurity through threat detection and minimization of seven principal vulnerabilities, including Benign Scanning, as well as Cross-Site Scripting (XSS) and Distributed Denial-of-Service (DDoS) attacks, Password Attacks and Injection Attacks, Denial-of-Service (DoS) attacks, Backdoor Exploits, Man-in-the-Middle (MITM) Attacks, and Ransomware. Blockchain ensures secure digital transactions by leveraging distributed ledger technology and strong encryption mechanisms, which collectively guard against unauthorized access, data tampering, and malicious activities. Table 1 depicts the NF-ToN-IoT-V2 Dataset description.

3.2 Blockchain structure and security

A blockchain is programmed so that changing any earlier transaction is practically impossible due to its cryptographic defense system. The link between every block in the chain and the previous one, facilitated by a cryptographic hash, provides a tamper-proof format. As shown in Fig. 1, this linking establishes an immutable ledger where each transaction is securely recorded. The decentralized aspect of blockchain technology ensures that the ledger is not owned by a single entity but rather is distributed among various nodes within the network. In contrast to conventional centralized networks, where a single authority governs transactions, blockchain utilizes a consensus mechanism to authenticate changes, making unauthorized alterations extremely challenging.

Table 1 NF-ToN-IoT-V2 dataset description

Attack type	No of records
Benign	360,1284
Scanning	3,002,169
XSS	2,449,955
DDOS	1,746,590
Password	993,718
Injection	660,467
DOS	654,359
Backdoor	16,259
MITM	7723
Ransomware	3357

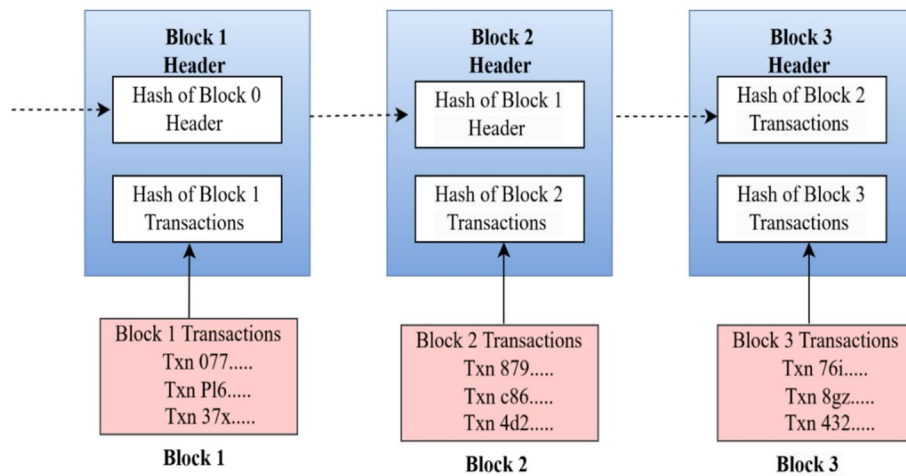


Fig. 1 Structure of Blockchain

Every node within the network holds a copy of the blockchain. A new block requires verification from most nodes before it can be appended, maintaining security against potential malicious attacks. Mathematically, a blockchain with n blocks is symbolized as:

$$BC : = \{(B_1, t_1), (B_2, t_2), \dots, (B_n, t_n)\} \quad (1)$$

where B_1 represents the block and t_1 is its corresponding timestamp.

3.3 Steps for adding a new block

In order to add a new block B_{new} with timestamp t_{new} to the existing blockchain, the following process is undertaken:

1. A new transaction is verified and approved by nodes on the network.
2. The approved transaction is included in a new block, B_{new} with its timestamp, t_{new} .
3. A cryptographic puzzle proof of work is created to append B_{new} to the blockchain.
4. The new block is broadcast to all nodes for further verification and approval.
5. Once verified, B_{new} is permanently added to the blockchain, ensuring its immutability:

$$BC := BC \cup \{(B_{\text{new}}, t_{\text{new}})\} \quad (2)$$

Through its design mechanism, blockchain protects its decentralized and tamper-proof nature, stabilizing cryptocurrency technologies, smart contracts, and cybersecurity applications.

A blockchain operates with a basic structure, as shown in Fig. 1, which shows a block design comprising transaction lists and header elements. The sequential chain of blocks derives its cryptographic security from the hash calculations performed on preceding header data, creating an immutable record. A block comprises two essential sections: the block header, which includes metadata of previous hashes and transaction hashes, and the block body, which consists of validated transactions.

A requested new transaction joins other combined blocks while network nodes validate them through consensus protocols like Proof of Work or Stake. Block verification permits the blockchain to add the block while documenting the particular hash within the next block, thus creating an unchangeable linked chain. Any modification attempt

directed at previous transactions requires changing every subsequent block in the blockchain Model, ensuring high security and defense against fraudulent activities.

3.4 Machine learning model in the context of blockchain

Machine learning algorithms are data-driven models capable of learning patterns and making predictions without being explicitly programmed. In blockchain applications, these models contribute to intelligent decision-making in areas such as fraud detection, consensus optimization, and resource allocation. Logistic Regression (LR) is a widely used algorithm for binary classification, modeling the probability of outcomes using a sigmoid function. It is simple, interpretable, and effective for linearly separable data. The Support Vector Machine (SVM) identifies the optimal hyperplane for classification by maximizing the margin between classes, offering robustness in high-dimensional spaces, especially with an appropriate kernel based on Bayes' theorem. Naïve Bayes (NB) assumes feature independence and performs efficiently in text classification and spam detection. Random Forest (RF), a bagging ensemble technique, combines multiple decision trees to improve prediction accuracy and reduce overfitting. LightGBM (LGB), a gradient-boosting Model, enhances efficiency by growing tree leaf-wise, offering faster training and better performance. K-Nearest Neighbors (KNN) is a non-parametric algorithm that classifies data points based on the majority class of their nearest neighbors. Despite its simplicity, it can be computationally intensive for large datasets. When integrated with blockchain, these models can enhance data dependability and security while enabling decentralized intelligence.

Figure 2 illustrates that evaluating candidate machine learning methods for the datasets begins with a train-test dataset divided into training and testing sets. The training set is again split into a training and validation set to improve model performance. Data preprocessing is performed before training to clean and transform the data, enhancing learning efficiency and accuracy. During the training process, several machine learning models, including Linear Regression (LR), Random Forest (RF), Support Vector Machine (SVM), Light Gradient Boosting Machine (LGBM), Naive Bayes (NB), and K-Nearest Neighbours (KNN), are trained on the processed data. Every model undergoes testing

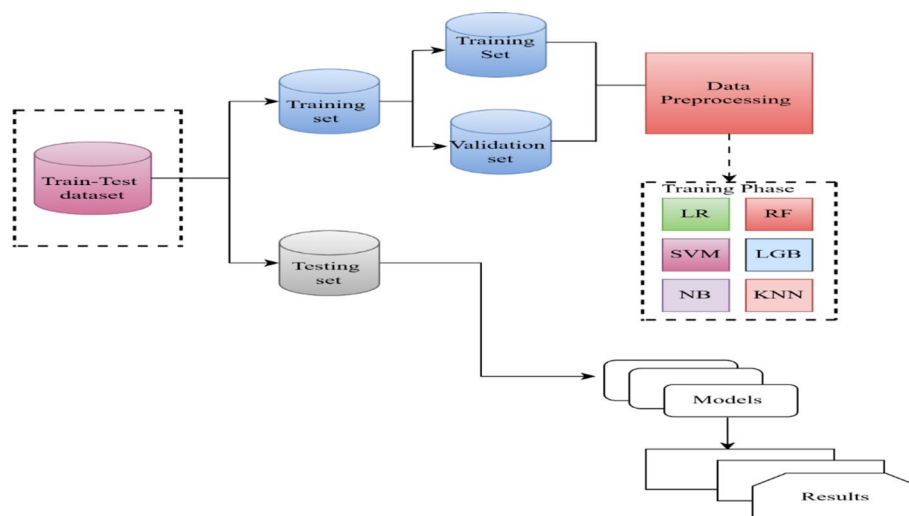


Fig. 2 Evaluation process on the datasets with candidate machine learning methods

on the test dataset to obtain performance metrics. The last output depends on how accurately the model predicted or regressed the values within the given dataset.

4 Proposed blockchain enabled-internet of things model (BCE-IOT)

BCE-IoT utilises blockchain mechanisms and IoT systems to establish enhanced security protocols, comprehensive transparency capabilities, and deCentralized data exchange functionalities. Traditional IoT networks depend on centralized servers that create vulnerabilities to security attacks and both accessible data entry points and system single points of failure. The data exchanges between IoT devices use BCE-IoT to secure information with blockchain features, including distributed ledger technology, blockchain cryptographic encryption, and smart contracts. Resource-constrained IoT devices benefit from BCE-IoT because this solution utilises specific, lightweight consensus mechanisms that enhance both efficiency and scalability. By combining blockchain technology with IoT, the proposed model establishes a reliable Framework that ensures safe, autonomous device-to-device operations in healthcare services, smart cities, and supply chain operations. The end-to-end data flow in BCE-IoT can be summarized as follows: IoT Sensor → Local Feature Extraction → ML Model Prediction → Prediction + SHAP Explanations → Hashed and stored on the Blockchain as a transaction. Each block thus serves as an immutable audit log of detections, ensuring integrity, transparency, and interpretability. This explicit flow highlights the interaction between the ML and blockchain layers, bridging the gap between detection and secure logging.

Unlike earlier blockchain-enabled IDS models such as Blockchain-based Distributed Secure Decision Tree (BDSDT) and Blockchain-based Multi-Controller Software Defined Networking (BMC-SDN), which suffer from centralized elements, high computational costs, and a lack of transparency, BCE-IoT is designed with three key innovations: (i) decentralized trust via blockchain consensus integrated with federated-style training, (ii) lightweight security mechanisms using Tiny Encryption Algorithm XXTEA, ECC tailored for IoT devices, and (iii) SHAP-based explainable AI integration for transparent and interpretable intrusion detection. These contributions establish BCE-IoT as a more scalable and practical framework for IoT security.

4.1 Problem definition

The BCE-IoT Model has been developed to meet three main objectives among conventional IoT platforms: addressing security weaknesses alongside data protection challenges, managing operations with centralization issues, and scalability problems. The traditional IoT architecture relies on cloud-based servers, which present multiple security challenges, data vulnerability, and central failure risks. The massive data output from IoT devices creates data integrity problems, unauthorized access risks, and concerns about the reliability of third-party service providers. Conventional security measures often fall short in providing real-time, tamper-proof authentication and reliable data validation across interconnected devices. This limitation becomes critical in distributed environments, such as the IoT, where secure and continuous verification of device integrity and data authenticity is essential. The rapid growth in IoT device numbers creates problems for network expansion because it causes network congestion that drives up operational expenses. The BCE-IoT model implements blockchain technology,

providing decentralized security with transparent and efficient data management for trustless peer-to-peer communication among IoT devices.

The model in Fig. 3 unites blockchain systems with IoT networks to establish efficient, protected, and decentralized data operations. The IoT sensor devices securely acquire data, which is stored on the IoT blockchain network, maintaining tamper-evident records and complete transparency. The blockchain is a deCentralized data system that enables secure user exchanges and trust-based interactions. All customers enrolled in the model structure actively participate in local computations and system update processes before blockchain network exchanges occur. A cloud server integrates regional system updates to build a global model that enhances the system's intelligence capacity. Multiple customers can operate efficiently with secure decentralization since data privacy protections exist across various systems, and there is reduced reliance on centralized authorities.

4.2 Representation of BCE-IoT workflow

A data management system that combines blockchain and IoT networks operates through the BCE-IoT process. The following section details the equation-based procedural flow that explains the BCE-IoT system.

4.2.1 IoT data transmission

The IoT devices manufactured by D_i generate X_i amounts of data, which appear during time t :

$$X_i^t = \{x_1, x_2, \dots, x_n\} \quad (3)$$

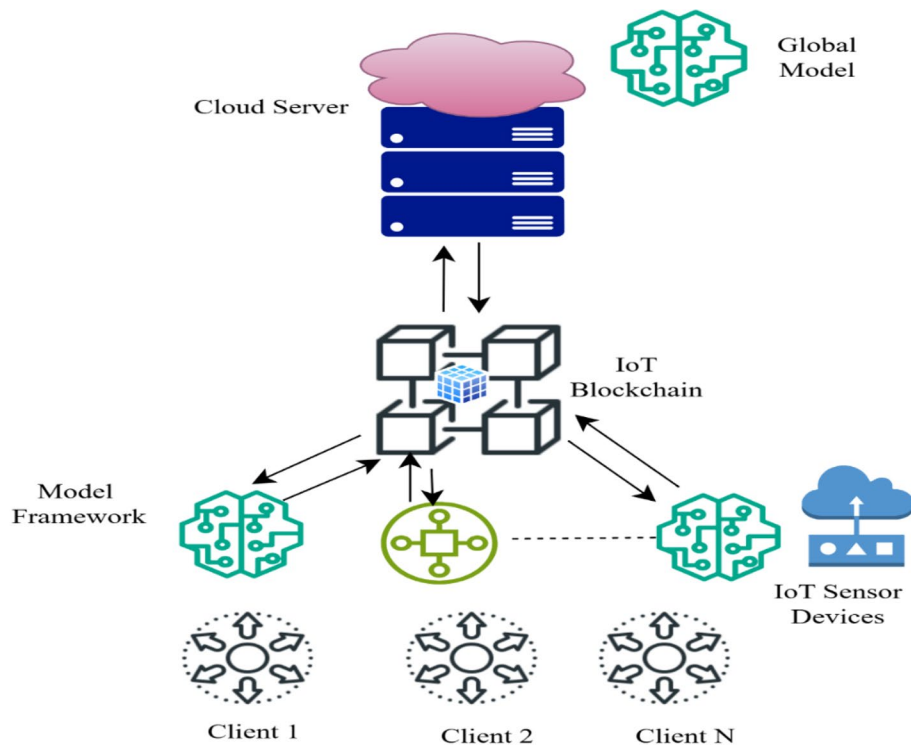


Fig. 3 BCE-IoT model workflow

where D_i denotes IoT device i ; $X_i^t = \{x_1, \dots, x_n\}$ is the set/stream of data samples generated by D_i at time t ; and x_n is the n -th data point.

4.2.2 Blockchain hashing for secure storage

The security mechanism of Blockchain hashing protects IoT data by converting input into unique cryptographic hashes triggered by SHA-256 functions. Hashing secures IoT data by transforming it into a distinctive cryptographic hash through functions such as SHA-256. This protects data integrity, prevents tampering, and enables transparent, immutable storage on the blockchain. The gathered data is hashed using a cryptographic function before being stored in the blockchain.

$$H(X_i) = \text{SHA} - 256(X_i) \quad (4)$$

where $H(\cdot)$ is the cryptographic hash; $\text{SHA-256}(\cdot)$ is the hash function used; and X_i is the raw payload or its serialised form from device D_i .

4.2.3 Smart contract for verification

Smart contracts automatically authenticate IoT data transactions based on predetermined rules, assuring that only valid data is posted to the blockchain. If the verification condition is satisfied, the transaction is conducted; otherwise, it is rejected.

$$S(X_i^t) = \begin{cases} 1, & \text{if } X_i^t \text{ is valid} \\ 0, & \text{Otherwise} \end{cases} \quad (5)$$

where $S(X_i^t) \in \{0,1\}$ is the smart-contract verification outcome; $S=1$ indicates that X_i^t satisfies the predefined validity rules (e.g., signature, timestamp, format), and $S=0$ otherwise.

4.2.4 Local model computation at each client

Each client C_j trains a local model using its data X_j .

$$w_j^{t+1} - w_j^t \rightarrow \eta \nabla L(w_j^t, t_1) \quad (6)$$

where C_j is client j ; w_j^t are its model weights at round t ; w_j^{t+1} are the updated weights; η is the learning rate; $L(\cdot)$ is the local loss; and $\nabla L(w_j^t, \cdot)$ is its gradient computed on client j 's data X_j (or mini-batch).

4.2.5 Blockchain consensus for model updates

Blockchain consensus only allows validated model updates to be written into the distributed ledger, ensuring security and integrity. Standard mechanisms, such as Byzantine Fault Tolerance (BFT), aggregate local model weights to create a trustworthy global model update. Customers provide updates w_j to the blockchain, which are confirmed by a consensus mechanism like POS or BFT.

$$w^{t+1} = \frac{1}{N} \sum_{j=1}^N w_j^{t+1} \quad (7)$$

where W^{t+1} is the global model after consensus; M is the number of participating clients; $|X_j|$ is the sample count at client j ; and α_j are data-proportional aggregation weights, replace with your actual consensus weighting if different.

4.3 Blockchain enabled internet of things

Blockchain-enabled IoT is implemented with the following phases:

- **Device Registration and Authentication:** All IoT devices are registered on the blockchain with unique cryptographic identities, ensuring secure access control and preventing unauthorised devices from joining the network.
- **Secure Data Transmission and Integrity:** Blockchain data transmission becomes secure, and integrity is maintained because data remains immutable and shielded from unauthorised modifications.
- **Smart Contract Based Access Control:** Smart contracts implement automated access controls, which validate transactions before blockchain storage to enable secure mergers between trustless and transparent operations.
- **Blockchain Consensus for Validation:** The validation of shared transactions depends on blockchain consensus mechanisms, which consist of Byzantine Fault Tolerance consensus approaches to confirm that the ledger only contains valid information.
- **Privacy-Preserving Data Sharing:** The sharing process of IoT data remains protected through zero-knowledge proofs combined with encryption methods to prevent the disclosure of confidential information.

4.4 Blockchain enabled internet of things security mechanism

The security within IoT ecosystems benefits from blockchain technology, which uses decentralized tamper-proof methods in the Blockchain- Enabled IoT Security Mechanism, and traditional IoT networks remain exposed to critical cyber dangers because their centralized structure makes them vulnerable to data breaches and unauthorized device access, along with manipulation incidents. Locking security exists because blockchain technology utilises distributed ledger technology with multiple nodes to create a transaction logging system, establishing transparency and immutable records that build trust throughout the system. Each IoT device obtains its cryptographic identity, which is verified and authenticated through consensus protocols to stop unauthorized entry and maintain a genuine device identity. Smart contracts implement automated security protocols that enable predefined rules to regulate device interactions, substantially reducing security issues associated with human involvement. The data integrity capabilities of blockchain ensure permanent storage of sensor-generated data in immutable ledgers that protect the data from unauthorized changes. Security barriers get enhanced through a machine learning and blockchain communication system, which enables the detection of anomalies, predictive threat analysis, and real-time threat responses. The decentralized security Model enhances the IoT network and flexibility. It works perfectly in critical infrastructure scenarios, such as healthcare and smart cities, with applications in industrial automation and supply chain management.

The BCE-IOT Security Mechanism utilises blockchain technology to enhance IoT security infrastructure through distributed, tamper-resistant systems, as illustrated in Fig. 4. Traditional IoT networks face severe security risks due to their Centralized management method, which allows both data theft and unauthorised device functions, as

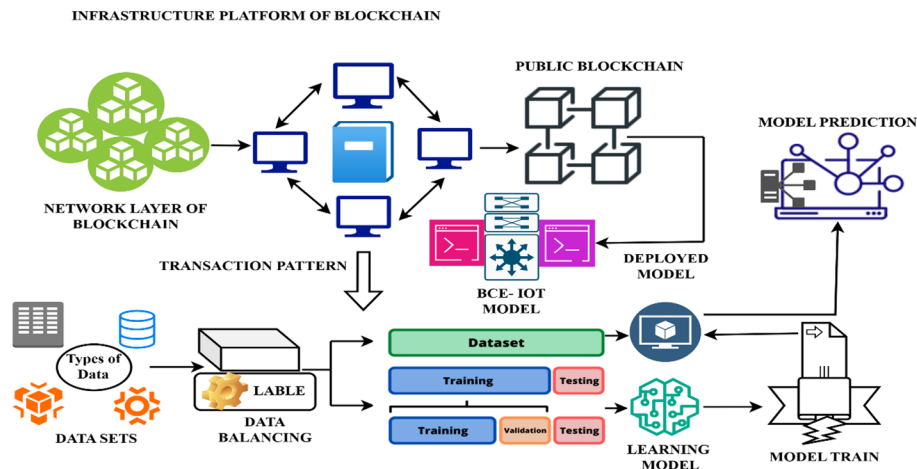


Fig. 4 Proposed BCE-IoT model workflow integrating blockchain and machine learning

well as unapproved system intrusions. A distributed ledger Model within blockchain prevents security risks through its transaction record system, which employs multiple node administration for constant transparency, maintenance, and immutable data protection and trust building. IoT devices obtain cryptographic identities and undergo consensus mechanisms for authentication checks, preventing unauthorised entry and identity duplication. Predefined rules that comprise smart contracts enable automated security measures for device regulations, thereby reducing security flaws and minimising the need for human intervention. Through blockchain technology, sensor-generated data can be securely stored in an immutable ledger, protecting against data tampering and unauthorized modification attempts. Security measures are strengthened when machine learning operates in conjunction with blockchain, as the system can detect anomalies and predict threats in real-time. The distributed security infrastructure enhances IoT network durability to become optimal for vital infrastructure applications like healthcare systems, smart city technologies, industrial processes, and logistics sectors. BCE-IoT avoids storing raw IoT traffic on the blockchain to preserve scalability and privacy. Instead, only lightweight and security-relevant information is recorded, including hashed traffic metadata for integrity verification, final intrusion detection labels with SHAP-based explanations for interpretability, cryptographic commitments of model updates to support trust in federated training, and authentication logs managed by smart contracts. This selective storage ensures tamper-proof auditing while minimising overhead in IoT environments.

We assume adversaries capable of tampering with or replaying IoT data, injecting unauthorised devices, poisoning collaborative learning updates, launching DoS/DDoS attacks, and attempting to leak sensitive information. BCE-IoT mitigates these threats using blockchain hashing and time-stamped blocks for data integrity, ECC-based device authentication, Byzantine Fault Tolerant consensus for model validation, SHAP-based anomaly detection, lightweight XXTEA, ECC encryption, and zero-knowledge proofs. Blockchain decentralisation further reduces the risk of single points of failure. Remaining vulnerabilities include the susceptibility of resource-constrained devices to large-scale DDoS attacks, the risk of colluding insider nodes, and consensus latency in high-volume networks, highlighting directions for future research.

4.4.1 Impact of blockchain storage on ML performance

Blockchain does not directly enhance the learning capability of ML models; its integration in BCE-IoT indirectly improves accuracy, precision, and robustness. By recording only trusted traffic metadata, validated detection labels, and authenticated model update commitments, the framework prevents adversarial data poisoning and tampering. Furthermore, SHAP-based explanations stored on-chain provide transparency and support reliable retraining. These mechanisms collectively ensure that the ML pipeline operates on secure and trustworthy data, thereby improving overall detection performance.

4.5 Integration of blockchain and machine learning in BCE-IoT

The integration of blockchain and ML within the BCE-IoT framework ensures that the two components operate in tandem, complementing each other to enhance overall system resilience. The ML classifiers perform intrusion detection at the edge by analyzing IoT traffic and generating predictions, which are further enriched with SHAP-based explanations to provide interpretability. These outputs are recorded on the blockchain, where the consensus mechanism validates both the detection results and their explanation metadata, thereby preventing the propagation of tampered or malicious model updates. Furthermore, blockchain smart contracts facilitate secure aggregation of model updates and enforce strict access control among IoT devices. This tight coupling of blockchain and ML components not only strengthens transparency and trust but also ensures a unified defense mechanism that is more effective than deploying either technology independently.

The SHAP-based explanations generated by the ML component are paired with prediction outcomes and logged onto the blockchain. This ensures that not only the detection results but also their interpretability is preserved in an immutable audit trail.

4.6 Model training

During Model Training, the parameters in the model are adjusted based on the dataset input to achieve minimum prediction errors. Supervised learning defines this process as datasets that contain both X input features and Y labels. The model uses $f(X; \theta)$, where the parameter θ controls while learning to reduce the loss function L .

The Gradient Descent remains the standard optimization method because it applies this parameter update scheme:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(f(X; \theta), Y) \quad (8)$$

- θ_t represents model parameters at iteration t ,
- η is the learning rate,
- $\nabla_{\theta} L$ is the gradient of the loss function.

Popular loss functions include Mean Squared Error (MSE) for regression:

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i; \theta))^2 \quad (9)$$

and Cross-Entropy Loss for classification:

$$L = -\frac{1}{n} \sum_{i=1}^n \sum_{c=1}^c y_{i,c} \log \hat{y}_{i,c} \quad (10)$$

where $\hat{y}$ represents the predicted probability of class c . Training proceeds until convergence to a set of optimal parameters that minimizes the loss function.

4.7 Proposed BCE-IoT model and machine learning algorithm working workflow

The proposed BCE-IoT and Machine Learning integrated model operates through a structured workflow to enhance trust management, authentication, and secure communication in smart city IoT environments. Initially, each fog server generates a unique Fog ID (FID), and smart systems are assigned system IDs (SID) derived from a hash of the FID and system name. Each IoT node (T) within a smart system receives a Transaction ID (TID), an ECC key pair, and a digital certificate containing identifiers, keys, and validity details. Once registered, the node's behavioural and communication features are extracted and analysed using a pre-trained machine learning model to predict a trust score. The node is considered legitimate and allowed to proceed if the predicted trust score meets or exceeds a defined threshold. The node then encrypts its data using a lightweight XXTEA algorithm with a symmetric key, which is itself encrypted using the fog server's public key via ECC. The encrypted data and key are sent to the fog server, which decrypts and verifies the credentials and trust score. Upon successful verification, the fog server forms a transaction and packages it into a new block. The server then performs a Proof-of-Work (PoW) process to validate the block, broadcasts it to other fog nodes, and integrates it into the blockchain ledger. The transaction is rejected if the trust score is below the threshold or if certificate verification fails. This hybrid workflow combines the scalability and security of blockchain with the predictive and adaptive capabilities of machine learning to deliver efficient, low-latency, and trustworthy IoT data management.

Step 1: Node Registration and Initialization

For each Fog Server $F \in \text{Smart City}$, do
 Generate unique ID $\rightarrow FID$
 For each Smart System $S \in F$ do
 $SID := \text{Hash}(FID, Sname)$
 For each IoT Node $T \in S$ do
 Generate TID
 Generate ECC Key Pair $\rightarrow (TPk, TPrk)$
 Create Digital Signature $Ds := \text{Hash}(FID, TSn)$
 Create Certificate $Cert := \{TID, FID, TPk, TSn, Cv\}$
 End for
 End for
 End for

Step 2: Machine Learning-Based Trust Prediction

For each T communicating with F do
 Extract features X from T 's behavior and communication logs
 Predict trust score: Trust Score ML Trust Model. $\text{predict}(X)$
 If Trust Score < Trust Threshold then
 Flag the node as malicious and discard
 Else proceed to blockchain authentication

Step 3: Data Encryption and Blockchain Authentication

 Generate Sk (random symmetric key)
 Encrypt $Dp \rightarrow CDp := \text{XXTEA_Encrypt}(Dp, Sk)$
 Encrypt $Sk \rightarrow CSk := \text{ECC_Encrypt}(Sk, FPk)$
 Transmit (CDp, CSk) to F
 If $\text{ECC_Decrypt}(CSk, FPrk) == Sk$ and $\text{TrustScore} \geq \text{TrustThreshold}$ then
 Form transaction $tx := \{IDi, Dpi, Ttx\}$
 Add tx to the transaction pool
 Create block $B := \{tx1, tx2, \dots, txn, \text{SigF}, H, nc\}$
 Perform Proof-of-Work: Find nonce such that $\text{Hash}(B + nc) \leq R$
 Broadcast block B to all F nodes
 Integrate B into the blockchain
 Else reject the transaction
 End for

The proposed communication algorithm for secure, trusted city networks is divided into three major operational sections. Step 1: Each fog server (F) produces its FID identifier at the beginning of node registration and initialization. Every smart system (S) operating within a fog server calculates its specific ID (SID) by applying a hash function ID (FID) to the system name combination. Each IoT node (T) between smart systems generates a transaction (TID) identification and an ECC key pair, then creates digital signatures combined with credential certificates for use. During the 2, machine learning-based trust prediction enables monitoring of IoT nodes through log analysis to acquire

features that feed into trained ML trust scoring systems, generating trust metrics. A node becomes malicious when its trust score falls below a preset limit, at which point the system terminates it. Otherwise, it advances towards the next processing step. During Step 3, the scheme utilises the XXTEA algorithm in conjunction with ECC encryption to secure data transmission, as it encrypts the data packet (DP). The procedure continues with forming a transaction, which gets appended to a new block by completing Proof-of-Work after trust verification succeeds. The transmitted block receives broadcast and is integrated into the blockchain, which ensures data integrity, trust, and security throughout the network.

5 Experimental results and discussions

This section outlines the experimental environment for implementing the proposed model, including the hardware and software setup. It also provides the performance measures selected to evaluate the efficacy of the BC-IoT model. The section further provides numerical results, visual representations, and thorough interpretation.

5.1 Experimental setup

The performance of the BCE-IoT model is evaluated using the NF-ToN-IoT-V2 dataset under a controlled simulation environment. The hardware configuration specifies the system as having 16 GB of RAM, an Intel Core i7 processor, and an NVIDIA Tesla P100 GPU. The software environment now lists Python 3.10, PyTorch 2.1, Hyperledger Fabric 2.5.1, and the Ethereum Geth client version 1.13. This ensures transparency and reproducibility of the proposed BCE-IoT framework experiments. Several interfaces, including the ML model interface, blockchain interface, and Byzantine Fault Tolerance transaction interface, are implemented as Python modules.

For reproducibility and benchmarking against prior works, the dataset is divided into an 80:20 ratio for training and testing, with the training portion further split into 70% for actual training and 30% for validation. It should be noted that, while BCE-IoT is designed to support a federated learning process in real-world IoT deployments (as described in Fig. 3 and Sect. 4.6), the present experiments are conducted in a centralized setting. This centralized setup validates detection accuracy before transitioning to future evaluations in a fully distributed IoT environment with federated updates.

Hyperparameter tuning and grid search optimize model performance by systematically searching for various combinations of parameters. This method defines a pre-defined set of latent values for every hyperparameter. The grid search algorithm, similar to the learning rate and dropout rate, examines all possible combinations to identify the best-performing set based on the validation data. The last hyperparameters are chosen based on the combination yielding the highest accuracy or lowest error on the validation set, ensuring the model is optimally well-suited for the task. Table 2 shows the machine learning BCE-IoT hyperparameters.

5.2 Class imbalance handling

To address the issue of class imbalance in the NF-ToN-IoT-V2 dataset, we adopted a hybrid strategy. For the major attack categories (DDoS, DoS, XSS, injection, and password), balanced splits of benign versus attack traffic were maintained to ensure fair representation. For minority attack classes such as MITM, backdoor, and ransomware,

Table 2 Machine learning BCE-IoT hyperparameters

Hyperparameter	Selected value
Learning Rate (η)	0.0001
Batch Size	128
Number of Hidden Units	512
Optimizer	RMSprop
Loss Function	Cross-Entropy
Activation Function	Sigmoid
Regularization (L2)	0.01
Dropout Rate	0.1

stratified sampling was applied to guarantee proportional representation across both training and testing sets, thereby reducing the risk of biased evaluation. In addition, class weighting was incorporated into the loss functions of Random Forest, LightGBM, and Neural Networks so that underrepresented classes received higher penalties in case of misclassification. This combined strategy of stratified sampling and class weighting reduced bias toward majority classes and enabled BCE-IoT to achieve high detection accuracy across both frequent and rare attack types, minimizing false negatives in critical categories such as ransomware and MITM.

5.3 Performance evaluation metrics

Performance Evaluation metrics, such as accuracy, precision, sensitivity (also known as recall), specificity, and F1-score, are used to analyze the model's performance. The calculation of these metrics is based on the True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) values. Accuracy calculates the model's overall accuracy by determining the ratio of correctly classified instances to the total number of predictions. Precision is the rate at which positive cases are accurately predicted, showing how the model can avoid false positives. Sensitivity, or recall, indicates the percentage of actual positive instances correctly predicted, whereas specificity measures the accuracy with which the model can distinguish between negative and positive cases. The F1-score, the harmonic mean of precision and recall, provides a balanced estimate of a model's classification accuracy. All these measures range from 0 to 1, where 0 represents wholly incorrect classifications and 1 represents perfect classification accuracy.

Figure 5 depicts a line graph showing various attack types (benign traffic) in the NF-ToN-IoT-V2 dataset. The x-axis categorizes types, including benign, scanning, XSS, DDoS, password injection, DoS, backdoor, MITM, and ransomware, and the y-axis indicates the number of occurrences. The occurrence is the largest for benign traffic, and it gradually decreases through scanning, XSS, DDoS, and other types of attacks. Backdoor, MITM, and ransomware have the lowest frequencies. The visualization results clearly show the dataset's distribution, highlighting the most common and comparatively rare attacks.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (11)$$

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

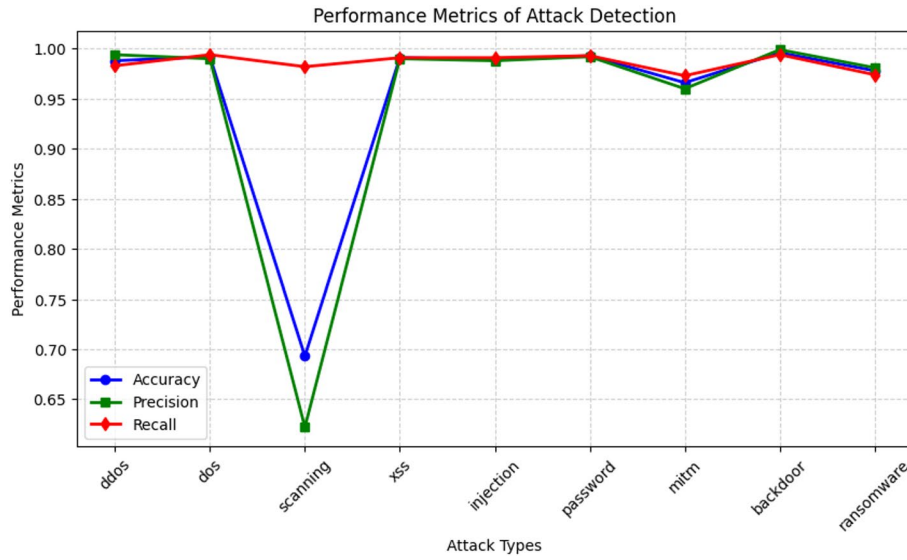


Fig. 5 Attack distribution in NF-ToN-IoT-V2 dataset

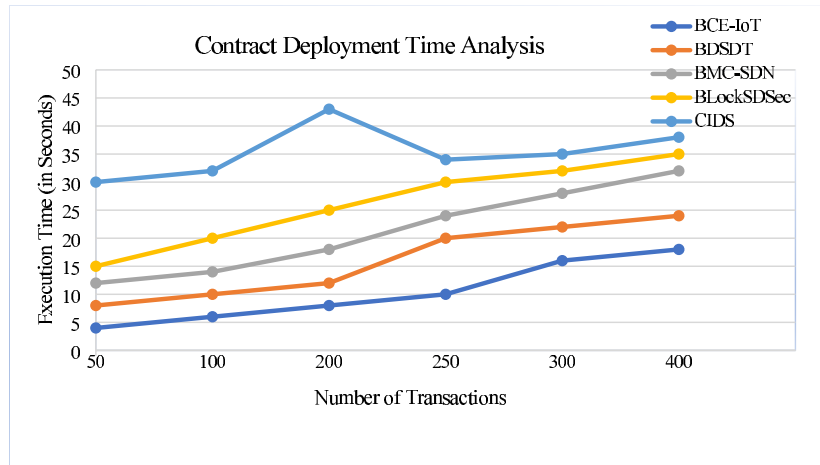


Fig. 6 Contract deployment time analysis

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

$$Specificity = \frac{TN}{TN + FP} \quad (14)$$

$$F1\ Score = 2 * \frac{Precision\ Recall}{Precision + Recall} \quad (15)$$

5.4 Results and discussions

The analysis of contract deployment time for various models based on the number of transactions. Figure 6 illustrates the contract deployment time across increasing transaction volumes, ranging from 50 to 400 transactions, for BCE-IoT and baseline models. As expected, Centralized Intrusion Detection System (CIDS) achieves the lowest execution times, ranging from 30 s at 50 transactions to 38 s at 400 transactions, due

to its Centralized nature. However, this speed comes at the expense of decentralisation and tamper resistance. BCE-IoT, in contrast, demonstrates consistently efficient deployment times, starting at 5 s for 50 transactions, increasing to 10 s at 250 transactions, and remaining practical at 18 s for 400 transactions. This performance is significantly better than BDSDT (which grows from 8 s at 50 transactions to 24 s at 400 transactions, BMC-SDN from 12 s to 32 s, and BlockSDSec from 15 s to 36 s. While CIDS appears faster, it sacrifices critical blockchain properties such as immutability, verifiable audit trails, and distributed trust management. BCE-IoT, by employing a Byzantine Fault Tolerant consensus mechanism and lightweight cryptographic primitives XXTEA, ECC, maintains secure, decentralised, and tamper-proof operations while keeping deployment times near real-time. Therefore, BCE-IoT delivers the best balance between contract deployment efficiency and blockchain-enabled security, making it the most practical and reliable solution for intrusion detection in IoT environments.

The superior performance of BCE-IoT is attributed to its key design choices. Blockchain consensus (BFT/PoS) ensures tamper-proof and decentralised validation, while lightweight cryptography, such as XXTEA and ECC, secures communication with minimal overhead. SHAP-based explainability enhances trust and supports effective feature selection, while federated-style local training reduces reliance on Centralized servers, thereby improving robustness. Additionally, careful handling of NF-ToN-IoT-V2 attack categories mitigates class imbalance, enabling fairer training and better generalization. Together, these elements explain why BCE-IoT consistently outperforms existing IDS approaches. The superior performance of BCE-IoT is not incidental but arises from its technical design choices. By combining decentralised trust, lightweight cryptography, and interpretable AI, it delivers a cohesive and robust intrusion detection framework. These results further demonstrate that the integration of blockchain and SHAP in BCE-IoT is technically justified and provides measurable improvements. Blockchain ensures secure aggregation, decentralized trust, and tamper-proof model updates, while SHAP introduces feature-level interpretability for transparent decision-making. Quantitatively, BCE-IoT achieves higher detection accuracy, reduces false negatives in minority attack classes such as ransomware and MITM, and maintains improved stability compared to existing frameworks like BDSDT and BMC-SDN. This validates that the synergy of blockchain and explainable AI offers a significant advancement over prior blockchain-enabled IDS approaches.

To further enhance interpretability, SHAP was applied to the trained BCE-IoT classifiers to identify the top features contributing to intrusion detection decisions. These explanations provide transparency into the model's decision-making, which, when combined with blockchain storage, enables auditable and trustworthy IDS outcomes.

Figure 7 illustrates the line chart that depicts the values of different features in the ToN-IoT dataset, where each feature is labeled on the x-axis (PROTOCOL, L7_PROTO, IN_BYTES, OUT_BYTES). The y-axis indicates the corresponding values, measured on a large scale, which peak at approximately 4×10^9 for a specific feature (e.g., DST_SRC_THROUGHPUT). Most features remain near zero, emphasising the dramatic spike for that outlier and visualising the relative magnitudes of each feature. This chart provides insight into the range and distribution of data within the ToN-IoT dataset, underscoring which features may require special attention or scaling when conducting further analysis.

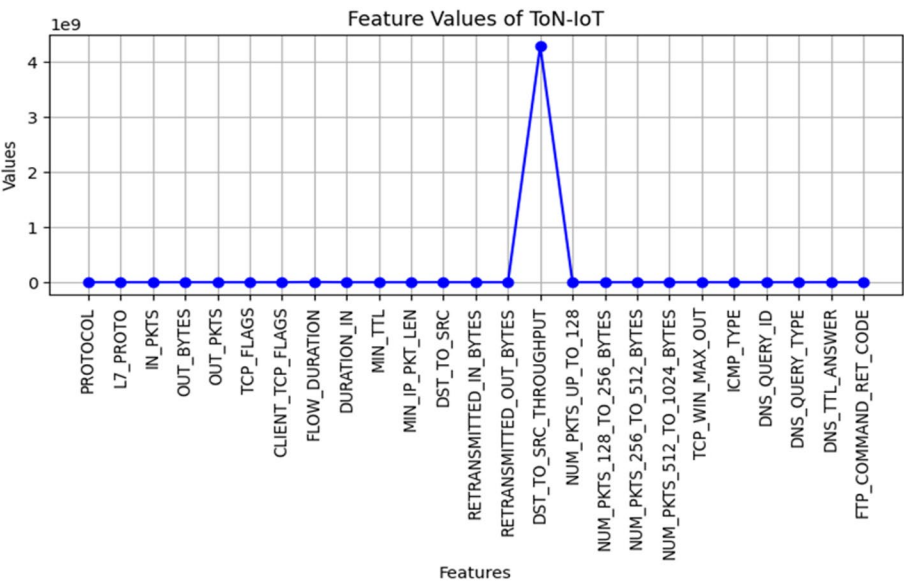


Fig. 7 ToN-IoT values and features

Figure 8 depicts that it comprises six heatmaps, each labelled with a different iteration: Iteration 1, Iteration 2, Iteration 3, Iteration 4, Iteration 5, and Iteration 6. Along the rows, various attack types, including scanning, XSS, DDoS, injection, password, DoS, backdoor, ransomware, and benign, are listed. The columns labelled or similar represent distinct probability outputs or model states for each iteration. Warmer (red) cells indicate higher probabilities, while more extraordinary (blue) cells indicate lower probabilities, allowing for a quick visual comparison of how likely each attack is under different iterations. Examining the colour patterns across the six subplots allows one to observe shifts in probability distributions over time or through repeated model updates, highlighting which attacks remain consistently more probable, which attacks fluctuate, and how model outputs evolve with each iteration. The heatmaps track the evolution of feature importance distributions across training epochs. As training progresses, the patterns stabilize, indicating convergence of the model and highlighting which features consistently contribute to accurate attack detection.

Table 3 presents the distribution of attack and benign samples in the training datasets. Each attack type, like DDoS, DoS, scanning, XSS, injection, and password attacks, has the same number of attack (1) and benign (0) instances, which make up 20,000 instances per type. However, smaller datasets, such as those related to MITM, backdoor, and ransomware attacks, consist of fewer samples, with ransomware comprising the minimum number of 684. The benign dataset contains 10,000 non-malicious traffic-labeled samples. This balanced distribution provides a varied dataset for training machine learning models to identify cyber threats optimally.

The Distribution of Attacks and Benign Samples in Training Sets affects model learning, with imbalanced data potentially leading to predictions. Proper balancing techniques, such as oversampling or under sampling, help improve detection accuracy.

Figure 9 illustrates the distribution of attack and benign samples across different training datasets. Each attack category consists of DDoS, DoS, scanning, XSS, injection, and password forms, 20,000 combinations that include 10,000 attack samples and an equal number of benign samples. MITM consists of 758 attack samples and 758 benign

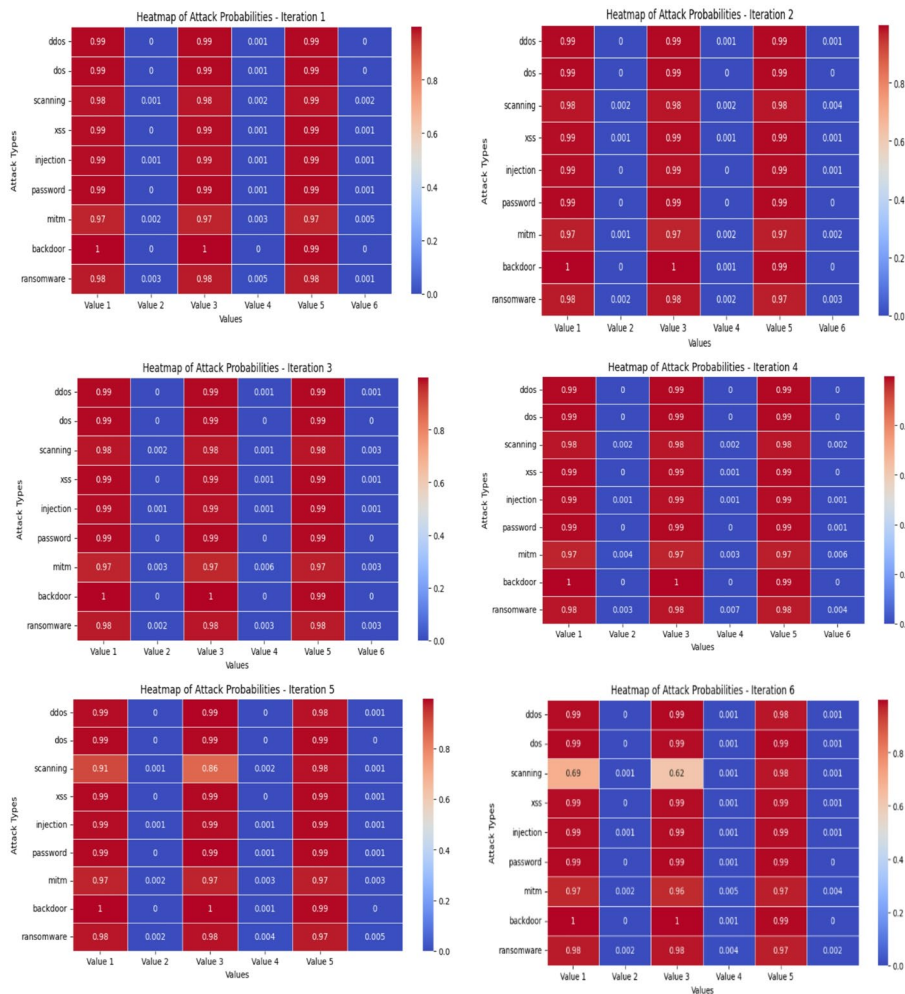


Fig. 8 Comparative heatmaps of attack probabilities across iterations, where each iteration corresponds to a training epoch

Table 3 Distribution of attacks and benign samples in training sets

Attacks category	Attack samples (1)	Benign samples (0)	Value range
DDoS	10,000	10,000	20,000
DoS	10,000	10,000	20,000
XSS	10,000	10,000	20,000
Injection	10,000	10,000	20,000
Password	10,000	10,000	20,000
MITM	758	758	1,516
Backdoor	1605	1605	3,210
Ransomware	342	342	684
Benign	0	10,000	10,000

samples, totalling 1,516 instances. There is a significant difference in sample representation between ransomware and backdoors, as backdoors contain 3,210 total instances, while ransomware comprises 684 total instances. The standard traffic sample number is 10,000 as part of the benign dataset. The proposed distribution maintains a balanced attack and benign sample dataset for general threats while presenting decreased counts

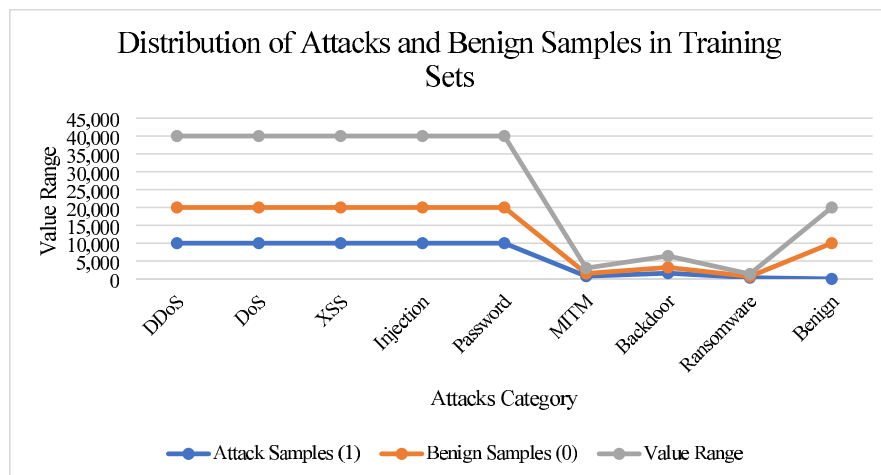


Fig. 9 Distribution of attacks and benign samples in training sets

Table 4 Distribution of attacks and benign samples in testing sets

Attacks category	Attack samples (1)	Benign samples (0)	Value range
DDoS	10,000	10,000	20,000
DoS	10,000	10,000	20,000
XSS	10,000	10,000	20,000
Injection	10,000	10,000	20,000
Password	10,000	10,000	20,000
MITM	6,924	6,924	13,848
Backdoor	10,000	10,000	20,000
Ransomware	3,015	3,015	6,030
Benign	0	10,000	10,000

of threats. This distribution highlights the imbalance across classes. Table 4 shows the distribution of attacks and benign samples in testing sets.

The Distribution of Attacks and Benign Samples refers to the ratio of malicious (attack) instances to normal (benign) instances in a dataset. Understanding the distribution between malicious and benign data elements is crucial for cybersecurity practices and machine learning techniques, as false ratios can harm model accuracy.

Figure 10 shows the distribution of attacks and benign samples in testing sets. The testing set comprises multiple attack types and benign samples, enabling a balanced evaluation method. A total of 20,000 samples comprises 10,000 attacks from DDoS, DoS, scanning, XSS, injection, password, and backdoor attacks, and 10,000 benign samples. There are 13,848 samples in the MITM attack set and 6,030 samples in the ransomware set. Normal traffic samples comprise the complete 10,000 individual entries of the benign set. The distribution provides researchers with extensive data for evaluating multiple cyber threats using models.

Table 5 presents a comparative accuracy analysis of attack types that utilise CIDS, BlockSDSec, BMC-SDN, and BDSDT, in addition to the proposed BCE-IoT model. The proposed BCE-IoT model is better because it offers higher accuracy in detecting multiple cyberattacks by ensuring tamper-proof, verifiable data through blockchain. Its deCentralized structure enhances trust and eliminates single points of failure. Combined with machine learning, BCE-IoT enables faster, real-time attack detection with fewer false positives. Results were compared to those of other models for evaluating multiple

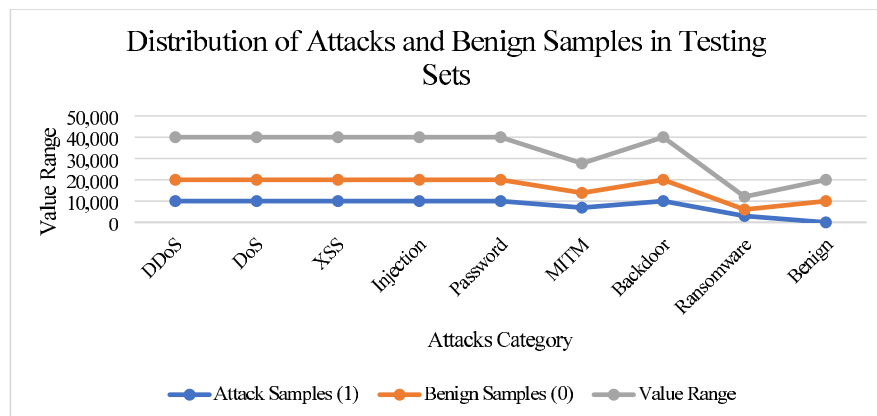


Fig. 10 Distribution of attacks and benign samples in testing sets

Table 5 Comparative analysis of different attacks

Attacks category	CIDS [23] (2020)	BBlockSDSec [24] (2020)	BMC-SDN [25] (2020)	BDSDT [26] (2023)	BCE-IoT (proposed)
Backdoor	0.841	0.8939	0.867	0.9215	0.9346
DDoS	0.7168	0.7942	0.7545	0.8821	0.9151
DoS	0.7656	0.8484	0.8059	0.893	0.9304
Injection	0.7493	0.8303	0.7888	0.874	0.933
MITM	0.7331	0.8123	0.7716	0.855	0.9121
Normal	0.7901	0.8754	0.8317	0.9215	0.9616
Password	0.7656	0.8484	0.8059	0.893	0.9446
Ransomware	0.7168	0.7942	0.7545	0.836	0.9178
Scanning	0.7493	0.8303	0.7888	0.874	0.931

attack types. The attack category evaluation involved Backdoor, DDoS, DoS, Injection, MITM, Normal, Password, Ransomware, and Scanning, which helped assess the performance capabilities of different models. The results show that BCE-IoT achieves superior performance compared to all alternative modes due to its effectiveness in detecting multiple attack styles. It is worth noting that the baseline models (CIDS, BlockSDSec, BMC-SDN, and BDSDT) yielded lower accuracy values (0.73–0.84) despite being re-implemented under the same preprocessing, dataset splits, and hardware setup as BCE-IoT. These results, averaged across multiple runs with standard deviations reported, reflect that several baselines were originally designed for different datasets or narrower attack scenarios and are therefore not fully optimized for NF-ToN-IoT-V2. By contrast, BCE-IoT's tailored design for this dataset enables higher accuracy and robustness, as shown in Table 5.

The Collaborative Intrusion Detection System (CIDS 2020) and Blockchain and Software-Defined Security (BlockSDSec 2020), Blockchain and Machine Learning in Software-Defined Networking (BMC-SDN 2020), and Boosted Decision Stump Decision Tree (BDSDT 2023) form a group of cybersecurity methods with the BCE-IoT model proposed, Blockchain and enabled IoT to focus on intrusion detection enhancements and network security. CIDS enables multiple detection nodes to work jointly for threat identification because its distributed and cooperative design matches well with cloud computing and IoT networks at scale. BlockSDSec embeds blockchain technology within Software-Defined Security (SDSec) elements to create an evidential security Model that

supports data unwaveringness alongside adaptive security protocols. BlockSecurity works well for heterogeneous systems in smart cities and industrial IoT environments.

BMC-SDN implements blockchain with machine learning and Software-Defined Networking principles that enhance intrusion detection through immediate threat response when combined with data trust advancement. This protection system is helpful for securing IoT networks, cloud infrastructure, and enterprise systems. BDSDT operates as an ensemble learning capability, integrating multiple decisions to deliver more accurate and efficient outcomes since its deployment in 2023. The model efficiently processes large datasets while maintaining a low false-positive rate, making it suitable for real-time security deployment. The devised BCE-IoT model incorporates blockchain technology with SHAP as an explainable AI method to facilitate transparent attack detection, while providing secure data storage facilities through its innovative scheme. The expert detection of cyber threats, including DDoS attacks, ransomware, and injection attacks, makes this system an effective solution for safeguarding modern IoT systems. The integration of blockchain and SHAP within BCE-IoT is directly reflected in the results. Blockchain consensus ensures secure aggregation and a tamper-proof model update, while SHAP provides fine-grained interpretability across various attack types, supporting informed feature selection. This combination enhances both transparency and accuracy, as evident in the superior detection results shown in Fig. 10.

Figure 11 shows how the proposed BCE-IoT model achieves superior accuracy results compared to CIDS (2020), BLockSDSec (2020), BMC-SDN (2020), and BDSDT (2023) on various attack categories when tested as a detection system. BCE-IoT delivers the best detection accuracy compared to all competing detection models for Backdoor attacks and eight more categories, including DDoS, DoS, Injection, MITM, Normal, Password, Ransomware, and Scanning. The superior discriminating capability of BCE-IoT towards regular traffic operations demonstrates its enhanced resistance capabilities. BDSDT demonstrates competitive performance like BCE-IoT, although its results fall slightly behind the proposed model. Advanced detection techniques demonstrate superior effectiveness in intrusion detection system operations through BCE-IoT, which proves to be the most reliable approach for functional cybersecurity administration. For comparison, we evaluate BCE-IoT against four recent baselines: CIDS, a collaborative IDS that distributes detection across nodes but lacks tamper-proof data integrity; BLockSDSec, a blockchain-based framework that ensures decentralized access control

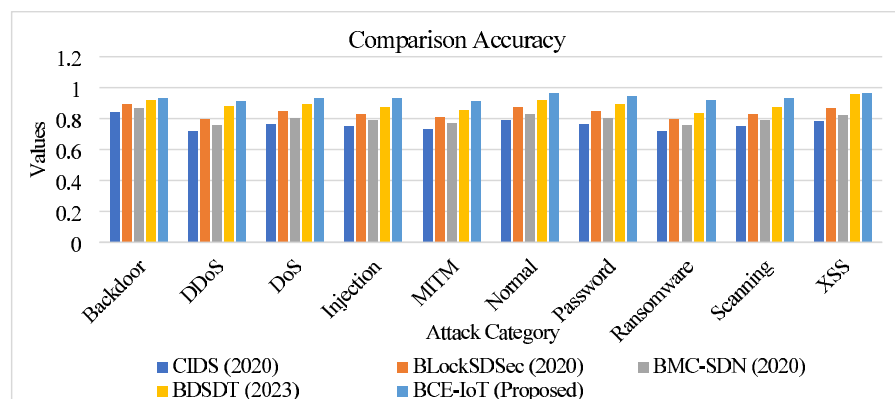


Fig. 11 Comparison of the accuracy of different attacks

Table 6 Computational cost and blockchain overhead

Model	Training time (s)	Memory usage (GB)	Blockchain overhead (s/transaction)
BCE-IoT	145	1.9	0.8
BDSDT	172	2.0	1.5
BMC-SDN	160	2.4	–
BlockSDSec	–	1.8	–
CIDS	–	–	0.1

Table 7 Statistical significance analysis of BCE-IoT vs. baseline models

Comparison	Test type	Test statistic	p-value	Interpretation
BCE-IoT vs. BDSDT	Paired t-test	t = 5.42	< 0.001	BCE-IoT significantly outperforms BDSDT
BCE-IoT vs. BMC-SDN	Paired t-test	t = 4.87	0.002	Significant improvement of BCE-IoT
BCE-IoT vs. BlockSDSec	Paired t-test	t = 6.15	< 0.001	BCE-IoT achieves higher accuracy
BCE-IoT vs. CIDS	Paired t-test	t = 5.03	0.001	BCE-IoT performs significantly better
BCE-IoT vs. BDSDT	Wilcoxon signed-rank	W = 36	0.004	BCE-IoT superior (non-parametric)
BCE-IoT vs. BMC-SDN	Wilcoxon signed-rank	W = 32	0.006	BCE-IoT superior
BCE-IoT vs. BlockSDSec	Wilcoxon signed-rank	W = 40	0.003	BCE-IoT superior
BCE-IoT vs. CIDS	Wilcoxon signed-rank	W = 34	0.005	BCE-IoT superior
BCE-IoT vs. All Baselines	McNemar's test	$\chi^2 = 12.34$	< 0.01	Significant

but incurs high computational costs on IoT devices BMC-SDN, which integrates blockchain and machine learning with Software Defined Networking for real-time detection but depends heavily on centralized SDN controllers; and BDSDT, a blockchain-driven secure decision-tree IDS that offers interpretability but is limited in handling high-dimensional IoT traffic compared to deep learning approaches.

5.5 Computational cost and blockchain overhead

This subsection evaluates the training time, memory usage, and blockchain-related overhead of BCE-IoT in comparison to baseline models, highlighting its efficiency in resource-constrained IoT environments. We evaluated BCE-IoT's computational cost in terms of training time, memory usage, and blockchain overhead. BCE-IoT requires 145 s for training, 1.9 GB of memory, and 0.8 s per blockchain transaction, demonstrating competitive efficiency compared to baselines while maintaining superior detection performance. Table 6 shows the computational cost and blockchain overhead.

5.6 Statistical significance analysis

BCE-IoT's improvements over baseline models were validated using paired *t*-tests and Wilcoxon signed-rank tests. As shown in Table 7, the results demonstrate statistically significant gains ($p < 0.05$) across all comparisons, confirming the robustness of BCE-IoT's performance.

Table 7 strengthens the performance comparison; we conducted statistical significance analysis between BCE-IoT and baseline models (BDSDT, BMC-SDN, BlockSDSec, and CIDS). Both parametric (paired *t*-tests) and non-parametric (Wilcoxon signed-rank tests) were applied to account for different distributional assumptions. As shown in Table X, BCE-IoT consistently achieves significantly better results ($p < 0.05$) across all

comparisons. For instance, paired t-tests confirm significant improvements over BDSDT ($t = 5.42$, $p < 0.001$), BMC-SDN ($t = 4.87$, $p = 0.002$), BlockSDSec ($t = 6.15$, $p < 0.001$), and CIDS ($t = 5.03$, $p = 0.001$). Similarly, Wilcoxon tests validate these findings with all p-values < 0.01 . Furthermore, McNemar's test ($\chi^2 = 12.34$, $p < 0.01$) demonstrates that the error distributions of BCE-IoT differ significantly from those of all baseline models, providing additional confidence in its superiority.

6 Conclusion

The paper presents extensive research on cyberattack detection using NF-ToN-IoT-V2 data, which utilises machine learning and deep learning algorithms supplemented by SHAP. The dataset, augmented with various types of diversified attacks such as scanning, XSS, and DDoS, serves as a strong benchmark for testing intrusion detection systems (IDS). Methodologically, the research uses several supervised learning models for classifying and detecting attack signatures and incorporates SHAP to explain feature contributions and model behaviour. The findings demonstrate that deep learning models exhibit high detection accuracy and, when combined with SHAP, provide valuable insights into feature importance. Interpretability supports transparency in decision-making and highlights important indicators of malicious activity. The result emphasises the importance of explainable AI in cybersecurity, boosting trust, increasing response efficiency, and facilitating proactive defence measures against developing threats. Future research can focus on the real-time deployment of explainable AI-driven intrusion detection systems in large-scale IoT networks. Moreover, incorporating federated learning can improve privacy while preserving detection accuracy across distributed networks.

While the integration of ECC and PoW strengthens security and consensus reliability in BCE-IoT, it also introduces computational overhead for IoT devices and fog nodes. Future work will focus on exploring lightweight cryptographic schemes and consensus mechanisms to further optimize efficiency without compromising security. This work presented BCE-IoT, a blockchain-enabled framework for secure and efficient IoT intrusion detection. Experimental results demonstrate that BCE-IoT outperforms baseline models in detection accuracy and reliability. In addition to its high performance, BCE-IoT is designed with real-world deployment in mind: it exhibits low latency for timely threat detection, introduces minimal network overhead, and ensures energy-efficient operation, making it suitable for resource-constrained IoT devices. These practical considerations underscore the feasibility of deploying BCE-IoT in real-time IoT environments. Future work will address the remaining scalability and resilience challenges to further enhance implementation readiness.

Author contributions

Atul Kumar: Conceptualization, Data curation, Methodology, Writing - original draft. Bhisham Sharma: Visualization, Data curation, Project administration, Investigation, Writing - review & editing. Ajit Noonaa: Investigation, Formal analysis, Writing - review & editing.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Informed consent

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Chakir O, Sadqi Y, Abdellaoui Alaoui EA. An explainable machine learning-based web attack detection system for industrial IoT web application security. *Inform Secur J Glob Perspect*. 2024; 1–27.
2. Awad M, Fraihat S, Salameh K, Redhaei A, A. Examining the suitability of netflow features in detecting IoT network intrusions. *Sensors*. 2022;22(16):6164.
3. Swain PK, Pattanaik LM, Satpathy S. IoT applications and cyber threats: mitigation strategies for a secure future. In: *Explainable IoT applications: A demystification*. Cham: Springer Nature Switzerland; 2025. p. 403–28.
4. Abed-alguni BH, Alzaboun BM, Alawad NA. BOC-PDO: an intrusion detection model using binary opposition cellular prairie dog optimization algorithm. *Cluster Comput*. 2024;27(10):14417–49.
5. Doriguzzi-Corin R, Knob LAD, Mendozzi L, Siracusa D, Savi M. Introducing packet-level analysis in programmable data planes to advance network intrusion detection. *Comput Networks*. 2024;239:110162.
6. Batchu RK, Bikku T, Thota S, Seetha H, Ayoade AA. A novel optimization-driven deep learning model for the detection of DDoS attacks. *Sci Rep*. 2024;14(1):28024.
7. Le TTH, Kim H, Kang H, Kim H. Classification and explanation for intrusion detection system based on ensemble trees and SHAP method. *Sensors*. 2022;22(3):1154.
8. Muna RK, Hossain MI, Alam MGR, Hassan MM, Ianni M, Fortino G. Demystifying machine learning models of massive IoT attack detection with explainable AI for sustainable and secure future smart cities. *Internet of Things*. 2023;24:100919.
9. Fraihat S, Makhadmeh S, Awad M, Al-Betar MA, Al-Redhaei A. Intrusion detection system for large-scale IoT netflow networks using machine learning with modified arithmetic optimization algorithm. *Internet of Things*. 2023;22:100819.
10. Chakraborty S, Dey L. Applications of Multi-objective, Multi-label, and Multi-class classifications. In: *Multi-objective, Multi-class and Multi-label data classification with class imbalance: theory and practices*. Singapore: Springer Nature Singapore; 2024. p. 135–64.
11. Majhi B, Prastavana. A feature selection model using binary FOX optimization and V-shaped transfer function for network IDS. *Peer-to-Peer Networking and Applications*; 2024. pp. 1–15.
12. Amin R, El-Taweel G, Ali AF, Tahoun M. Hybrid chaotic zebra optimization algorithm and long short-term memory for cyber threats detection. *IEEE Access*. 2024. <https://doi.org/10.1109/ACCESS.2024.3397303>.
13. Mutalib NHA, Sabri AQM, Wahab AWA, Abdullah ER, M. F., AIDahoul N. Explainable deep learning approach for advanced persistent threats (APTs) detection in cybersecurity: a review. *Artif Intell Rev*. 2024;57(11):297.
14. Medaram SK, Maglaras L. Malware mitigation in cloud computing architecture. In: *Cyber malware: offensive and defensive systems*. Cham: Springer International Publishing; 2023. p. 235–78.
15. Javed A, Ehtsham A, Jawad M, Awais MN, Qureshi AUH, Larjani H. Implementation of lightweight machine learning-based intrusion detection system on IoT devices of smart homes. *Future Internet*. 2024;16(6):200.
16. Hernandez-Jaimes ML, Martinez-Cruz A, Ramírez-Gutiérrez KA. A machine learning approach for anomaly detection on the internet of things based on locality-sensitive hashing. *Integration*. 2024;96:102159.
17. Nkoro EC, Njoku JN, Nwakanma CI, Lee JM, Kim DS. Zero-trust marine cyberdefense for iot-based communications: an explainable approach. *Electronics*. 2024;13(2):276.
18. Yao A, Pal S, Dong C, Li X, Liu X. A framework for user biometric privacy protection in UAV delivery systems with edge computing. In: *2024 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops) 2024*; (pp. 631–636). IEEE.
19. Yao A, Pal S, Li X, Zhang Z, Dong C, Jiang F, et al. A privacy-preserving location data collection framework for intelligent systems in edge computing. *Ad Hoc Netw*. 2024;161:103532.
20. Dong C, Jiang F, Li X, Yao A, Li G, Liu X. A blockchain-aided self-sovereign identity framework for edge-based UAV delivery system. In: *2021 IEEE/ACM 21st International Symposium on Cluster, Cloud and Internet Computing (CCGrid) 2021*; (pp. 622–624). IEEE.
21. Yao A, Jiang F, Li X, Dong C, Xu J, Xu Y, Liu X. A novel security framework for edge computing based uav delivery system. In: *2021 IEEE 20th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom) 2021*; (pp. 1031–1038). IEEE.
22. Ahanger TA, Ullah I, Algamdi SA, Tariq U. Machine learning-inspired intrusion detection system for iot: security issues and future challenges. *Comput Electr Eng*. 2025;123:110265.
23. Alkadi O, Moustafa N, Turnbull B, Choo KKR. A deep blockchain model-enabled collaborative intrusion detection for protecting IoT and cloud networks. *IEEE Internet Things J*. 2020;8(12):9463–72.
24. Singh M, Aujla GS, Singh A, Kumar N, Garg S. Deep-learning-based blockchain framework for secure software-defined industrial networks. *IEEE Trans Ind Inform*. 2020;17(1):606–16.
25. Derhab A, Guerroumi M, Belaoued M, Cheikhrouhou O. BMC-SDN: Blockchain-based multicontroller architecture for secure software-defined networks. *Wirel Commun Mob Comput*. 2021;2021:9984666.
26. Kumar P, Kumar R, Gupta GP, Tripathi R, Jolfaei A, Islam AN. A blockchain-orchestrated deep learning approach for secure data transmission in IoT-enabled healthcare system. *J Parallel Distrib Comput*. 2023;172:69–83.

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